

Solution Requirements (Functional & Non-functional)

Date: 31 January 2025 **Team ID:** SWTID-2026-7464 **Project Name:** Script-Controlled ACL – Restrict Record Access
Based on Field Value **Maximum Marks:** 4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	ACL Configuration	Configure an ACL for the target record and identify the field whose value controls record access.
		Define the expected field value or condition that determines whether access is allowed or denied.
FR-2	Script-Controlled Access Evaluation	Execute the ACL script when a user requests a record and evaluate the configured field-value condition.
		Return an allow/deny decision based on the field-value condition.
FR-3	Record Access Enforcement	Allow authorized users to access records when the field value satisfies the ACL condition.
		Restrict access when the field value does not satisfy the configured condition.
FR-4	Validation & Exception Handling	Handle missing, empty, invalid, or unexpected field values using defined access behavior.
		Validate the ACL logic with different field values and user contexts before deployment.
FR-5	Audit & Maintenance	Record or document ACL decisions and test outcomes where required for troubleshooting.
		Allow administrators to update the field-based rule when the business access requirements change.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	ACL configuration and field-value conditions should be clear enough for administrators to manage.
NFR-2	Security	Access decisions must enforce the configured field-based restriction consistently and protect sensitive data.
NFR-3	Reliability	The ACL script should produce consistent allow/deny decisions for the same record, field, and user.
NFR-4	Performance	Field-value evaluation should add minimal overhead to record-access requests and avoid bottlenecks.
NFR-5	Availability	The access-control logic should remain available whenever the protected records are accessed.
NFR-6	Scalability	The solution should support applying the field-based ACL pattern to additional records, fields, and users.

Project scope note: The requirements above are project-specific adaptations of the uploaded template. They focus on restricting record access based on a record field value using script-controlled ACL logic.